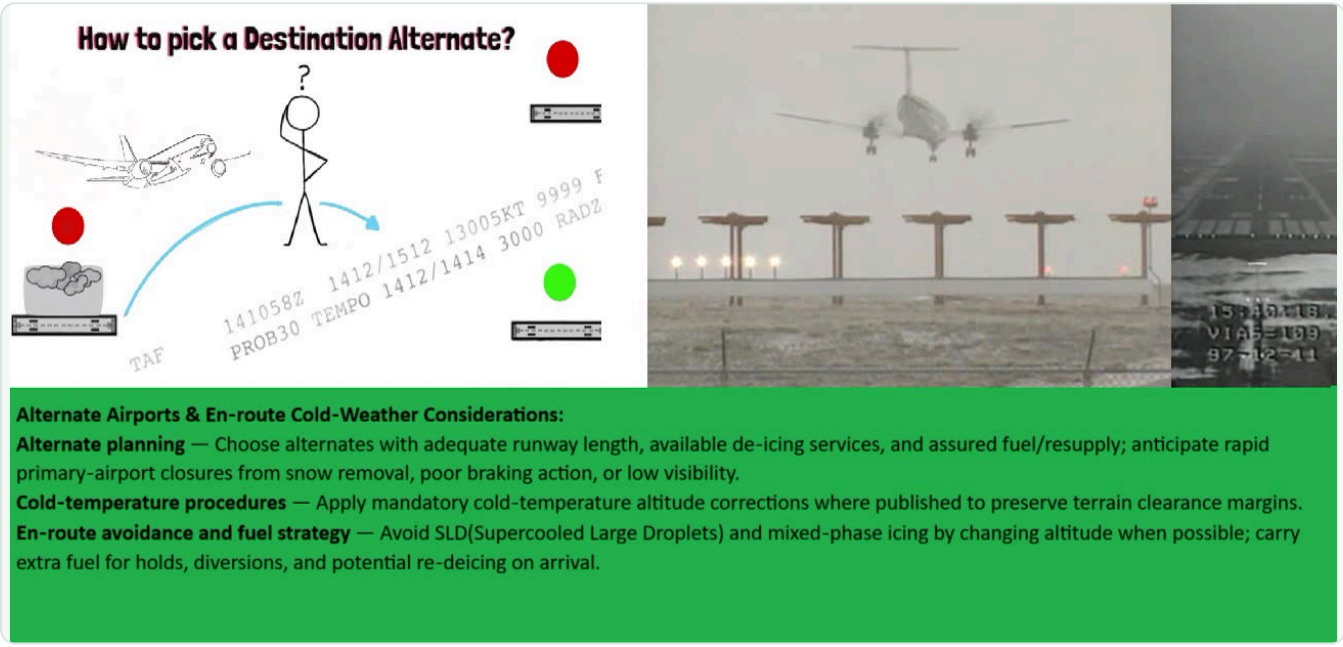
MODULE 6 · OPERATIONS AND HUMAN FACTORS

Lesson 24: Cold Temperature Altimetry and En-route Considerations

Lesson 24 of 30 · about 9 min

LEARNING OBJECTIVES

- Explain why a barometric altimeter over-reads in cold air.
- Apply cold temperature corrections to minimum altitudes.
- Recognise the limits of baro-VNAV approaches at low temperature.



Choose alternates that are not affected by the same winter weather.

Why cold air is dangerous for terrain clearance

A barometric altimeter is calibrated for the ICAO Standard Atmosphere (ISA). In air colder than ISA, the pressure levels are closer together. The altimeter then shows a higher altitude than the true altitude. **The aircraft is lower than indicated** — “from high to low, look out below”.

The error is zero at the altimeter setting source (the aerodrome) and grows with height above it. It is greatest in very cold air and over high terrain.

Rule of thumb

> **APPROXIMATION**
> Correction ≈ **4 % of the height above the aerodrome for every 10 °C below ISA** (ISA deviation measured at the aerodrome).
>
> Example: aerodrome elevation 2 300 ft, temperature −20 °C. ISA at 2 300 ft ≈ 15 − 4.6 ≈ +10 °C, so ISA −30 °C. Initial approach fix at 5 300 ft, which is 3 000 ft above the aerodrome. Correction ≈ 3 × 4 % × 3 000 = 360 ft. Fly 5 660 ft, or round it up as the company procedure requires.

For accurate values, use the ICAO PANS-OPS cold temperature correction table or the company table. They apply to the published altitudes on the approach: minimum sector, initial, intermediate and final approach fix altitudes, step-down fixes, missed approach altitudes and the DA/MDA.

Applying corrections

- The pilot is responsible for applying cold temperature corrections to minimum altitudes, unless the Operations Manual states otherwise.
- **Tell ATC** when you apply corrections to an assigned or published altitude, because it affects separation.
- Do not correct altitudes assigned by radar controllers without coordination. Ask whether the assigned altitude is temperature-corrected if you are near terrain.
- The correction applies to the **altimeter** altitudes, not to radio altimeter heights (e.g. a CAT II/III decision height).

Baro-VNAV approaches

A baro-VNAV glide path is affected by the same error: in cold air the actual path is **lower and shallower** than published. RNP approach charts with LNAV/VNAV minima state a **minimum temperature** below which baro-VNAV must not be used. Some FMS can apply temperature compensation. Use it only as your FCOM and Operations Manual permit.

En-route winter considerations

- Very cold upper air masses: watch the fuel temperature (Lesson 22).
- Minimum en-route altitudes over high terrain need cold temperature awareness, especially at night with weak pressure gradients.
- Alternates may be affected by the same snow, icing or freezing rain as the destination. Check runway condition forecasts and de-icing availability.

KEY TAKEAWAYS

- In cold air the aircraft is lower than the altimeter shows.
- Correction ≈ 4 % per 10 °C below ISA per height above the aerodrome — use the official table.
- Correct minimum and approach altitudes and inform ATC; respect baro-VNAV temperature limits.

KNOWLEDGE CHECK

Aerodrome ISA deviation is −20 °C. A fix is 2 000 ft above the aerodrome. Using the rule of thumb, what is the approximate correction?

A. About 40 ft

B. About 160 ft

C. About 400 ft

D. About 800 ft

Answer: B. 2 × 4 % = 8 %; 8 % of 2 000 ft = 160 ft. Add it to the published altitude.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.